

Xiaohan Zhang

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RESEARCH SUMMARY

Robot Learning and Planning

My goal is to develop learning algorithms that use perception to guide long-horizon planning for autonomous robots. My research uses techniques from robot planning, including probabilistic planning, task and motion planning, and machine learning, such as deep learning, reinforcement learning.

ACADEMIC BACKGROUND

State University of New York at Binghamton, Binghamton, NY

Ph.D. in Computer Science

Aug. 2020 – Present

- Advisor: Shiqi Zhang

State University of New York at Binghamton, Binghamton, NY

M.S. in Computer Science

Aug. 2019 – May 2020

Renmin University of China, Beijing, China

B.E. in Computer Science and Technology

Sep. 2015 – May 2019

EMPLOYMENT

Google Brain, Robotics, Mountain View, CA

Student Researcher

Mar. 2022 - Aug. 2022

- Advisors: Montserrat Gonzalez and Fei Xia

Autonomous Intelligent Robotics Lab, SUNY Binghamton, Binghamton, NY

Research Assistant

Dec. 2020 - Present

- Collaborators: Peter Stone, Yuke Zhu, Jesse Thomason, and Jivko Sinapov

Department of Computer Science, SUNY Binghamton, Binghamton, NY

Teaching Assistant

Jan. 2020 - Dec. 2020

- CS 465/565: Intro to Artificial Intelligence

PREPRINTS

1. Thomas Lew, Sumeet Singh, Mario Prats, Jeffrey Bingham, Jonathan Weisz, Benjie Holson, **Xiaohan Zhang**, Vikas Sindhwani, Yao Lu, Fei Xia, Peng Xu, Tingnan Zhang, Jie Tan, and Montserrat Gonzalez. Robotic Table Wiping via Reinforcement Learning and Whole-body Trajectory Optimization. *Under review*.
2. Yan Ding, **Xiaohan Zhang**, Saeid Amiri, Nieqing Cao, Hao Yang, Chad Es-selink, and Shiqi Zhang. Robot Task Planning and Situation Handling in Open Worlds. *Under review*.
3. Yan Ding, Cheng Cui, **Xiaohan Zhang**, and Shiqi Zhang. GLAD: Grounded Layered Autonomous Driving for Complex Service Tasks. *Under review*.

JOURNAL ARTICLES

1. **Xiaohan Zhang**, Saeid Amiri, Jivko Sinapov, Jesse Thomason, Peter Stone, and Shiqi Zhang. Multimodal Embodied Attribute Learning by Robots for Object-Centric Action Policies. *Submitted to Autonomous Robots, 2022*

2. Yan Ding, **Xiaohan Zhang**, Xingyue Zhan, and Shiqi Zhang. Learning to Ground Objects for Robot Task and Motion Planning. *IEEE Robotics and Automation Letters (RA-L)*, 2022

CONFERENCE PAPERS

1. **Xiaohan Zhang**, Yifeng Zhu, Yan Ding, Yuke Zhu, Peter Stone, and Shiqi Zhang. Visually Grounded Task and Motion Planning for Mobile Manipulation. *IEEE International Conference on Robotics and Automation (ICRA)*, 2022
2. **Xiaohan Zhang**, Jivko Sinapov, and Shiqi Zhang. Planning Multimodal Exploratory Actions for Online Robot Attribute Learning. *Robotics: Science and Systems (RSS)*, 2021
3. Kishan Chandan, Jack Albertson, **Xiaohan Zhang**, Xiaoyang Zhang, Yao Liu, and Shiqi Zhang. Learning to Guide Human Attention on Mobile Telepresence Robots with 360 Vision. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2021
4. Yan Ding, **Xiaohan Zhang**, Xingyue Zhan, and Shiqi Zhang. Task-Motion Planning for Safe and Efficient Urban Driving. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2020

WORKSHOP PAPERS

1. **Xiaohan Zhang**, Jivko Sinapov, and Shiqi Zhang. (Extended Abstract) Information-Theoretic Reward Shaping for Multimodal Object Attribute Learning. *Workshop on Integrating Planning and Learning, Robotics: Science and Systems (RSS Workshop)*, 2021
2. Yan Ding, **Xiaohan Zhang**, Xingyue Zhan, and Shiqi Zhang. (Extended Abstract) Task-Motion Planning for Safe and Efficient Urban Driving. *Workshop on Interaction and Decision-Making in Autonomous Driving, Robotics: Science and Systems (RSS Workshop)*, 2020
3. Kishan Chandan, **Xiaohan Zhang**, Jack Albertson, Xiaoyang Zhang, Yao Liu, and Shiqi Zhang. Guided 360-Degree Visual Perception for Mobile Telepresence Robots. *Workshop on Closing the Academia to Real-World Gap in Service Robotics, Robotics: Science and Systems (RSS Workshop)*, 2020

ACADEMIC SERVICES

Program Committee Member:

- AAAI Conference on Artificial Intelligence (AAAI), 2023

Journal Reviewer:

- IEEE Robotics and Automation Letters (RA-L), 2022

Conference Reviewer:

- IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2021, 2022

TECHNICAL SKILLS

Programming Languages: Python, C/C++, ASP (Answer Set Programming), PDDL (Planning Domain Definition Language), Java, JavaScript, HTML
Softwares and Tools: ROS, Gazebo, CARLA, Unity, AI2-THOR, iGibson, Tensorflow, PyTorch, Clingo, Git, Vim, LaTeX